POP N LOCK

Sky Wolf Studio

II

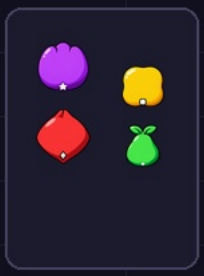
YOUR PLOT 12,480

TOYZ
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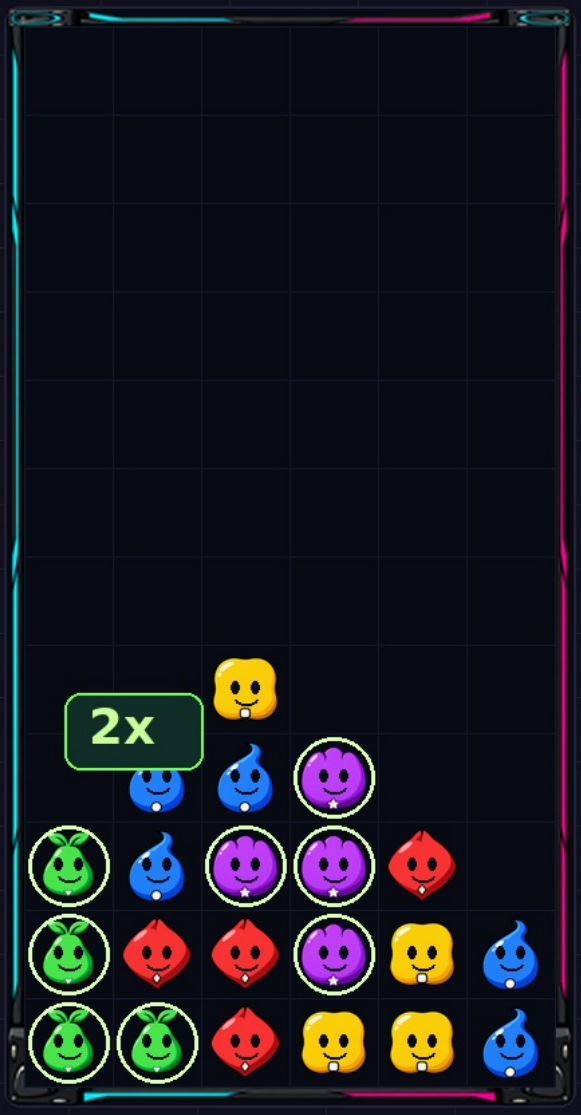
INCOMING
6

NEXT



HYPE

FRESH



A chain finds its rhythm

Cancelled 6 Chaff



LEFT | SOFT | RIGHT

ROTATE L | ROTATE R | DROP

Pop N Lock Review

Sky Wolf Studio | 24 October 2026

Priority 1: core release gate. Priority 2: next complete feature slice. Priority 3: later expansion. Effort S: localized change; M: connected subsystem; L: multiple systems and substantial validation. Estimates exclude art production.

Core gameplay

1. Fix the feel before expanding the game

What and why
First ship the verified fixes, dependable input and readable boards. Then add progression.
This is a review of the live v1.6 page and its downloaded source, not a deployed update.
Touch latency, mobile browser bars and expert difficulty still require physical device tests.

Where on screen Title, Campaign, How to play and match screens; source snapshot dated 23 October 2026.

Effort M | Priority 1 | PNL-001

2. Separate rules from controls

What and why
Keep the current countering rules as Standard; offer Retro without countering or All Clear.
SEGA identifies countering and quick turn as additions in Puyo Puyo Tsu. See sources.json.
Keep Powers separate. Store ruleset, input profile and difficulty in every replay and record.

Where on screen Title > Play options; show Standard, Retro or Powers beside the match score.

Effort M | Priority 1 | PNL-002

3. A small landing window

What and why
Start a 300 ms grounded timer on first contact, with at most two successful move resets.
Cap total contact time at 650 ms; hard drop locks immediately. Values are tuning proposals.
Compare against the exact cartridge version the director chooses before calling this authentic.

Where on screen Player board; gravityStep, softStep and lockPiece.

Effort M | Priority 1 | PNL-003

4. Deliberate rotation and quick turn

What and why
Test rotation at both walls, on ledges, beneath roofs and inside a one column shaft.
Use a documented kick table; add a two press 180 degree quick turn within 250 ms in Standard.
Never teleport through occupied cells. Keep collision state immediate and the 90 ms tween visual.

Where on screen Both rotation controls; tryRotate, fits and the pivot marker in the falling pair.

Effort M | Priority 1 | PNL-004

5. One predictable repeat scheduler

What and why
Keep 105 ms delay and 33 ms repeat as the Fast preset; offer 160 ms and 50 ms for Relaxed.
Use simulation time for keyboard and touch alike; the first repeat must occur at the delay.
Allow one rotation buffered for 120 ms across spawn; never buffer a hard drop across pieces.

Where on screen Settings > Controls; c-left, c-right and keyboard handlers.

Effort M | Priority 1 | PNL-005

6. Separate movement from drop intent

What and why
Default to buttons and soft drop. Make the upward hard drop flick an explicit optional setting.
Lock a drag to horizontal or vertical intent after 12 CSS px, using a 1.5 axis ratio.
Use pointer capture and clear every hold on cancel, blur, pause, visibility loss and match end.

Where on screen Player board gesture surface and bottom control tray.

Effort M | Priority 1 | PNL-006

7. Bring back a visible soft drop

What and why
The live tray has five buttons and no c-down element, although a soft hold handler exists.
Use two rows: left, soft drop, right; then rotate left, rotate right, Drop. All targets 56 px.
Allow a mirrored thumb layout. Keep Drop visually separated and always require fresh intent.

Where on screen Match > bottom tray; Settings > Controls.

Effort M | Priority 1 | PNL-007

8. Use the phone height without shrinking text

What and why
At 412 wide, propose a 264 x 528 player grid and a 96 x 192 rival grid in a separate rail.
At 360 and 375 wide use 228 x 456 and 240 x 480 player grids; retain 48 px minimum controls.
Use visualViewport and safe areas. Put labels in CSS pixels, with a 12 px practical minimum.

Where on screen Match layout; retain the existing visualViewport measurement instead of innerHeight alone.

Effort M | Priority 1 | PNL-008

9. Keep the landing guide accurate

What and why
Show both destination cells including split landings; a faint connector marks the active pivot.
Resolve each disconnected pod after contact. Never let the guide promise an impossible cell.
Provide Off, Outline and Solid guide options; keep markings visible in every pod and preview.

Where on screen Player well and next pair window; ghostY and split pair settling.

Effort M | Priority 1 | PNL-009

10. A readable chain rhythm

What and why
Tuning start: 80 ms landing squash, 230 ms pop, 110 to 180 ms settle, then the next scan.
Calculate landing from distance and animate to the logical cells; no frame dependent decay.
Play input sounds immediately. Music beats can follow clears but must never delay placement.

Where on screen Both wells; resolveTick, decayFall and the sound dispatch.

Effort M | Priority 1 | PNL-010

11. Make every attack explainable

What and why
Display queued Chaff as a number plus row icons, with a separate cancelled amount.
The 30 cap limits landing per turn, not the attack sent. Preserve all remaining queued Chaff.
Commit both fields on the same simulation tick before evaluating victory or mutual top out.

Where on screen Above each board; finishResolve, dropPending and endMatch.

Effort M | Priority 1 | PNL-011

12. Offline comfort and fair interruption

What and why
Pause Solo and Campaign when the app hides; save board, queue, seed and clocks atomically.
Resume through a three count. Online uses a published reconnect rule instead of silent pausing.
The present watchdog advances every running mode in the background; gate it by mode.

Where on screen Pause sheet, Android app switching and page visibility handler.

Effort M | Priority 1 | PNL-012

Powers

13. Replace control sabotage with board tactics

What and why
Cut input reversal, rotation lock, forced drop, hidden colours and full control freeze.
The supplied power design contradicts itself on offset bypass and safe board mutations.
Use one queued power per field, cancelable Chaff attacks, and no active pair mutations.

Where on screen Powers Campaign loadout and cast preview; original design copied in extras for comparison.

Effort L | Priority 2 | PNL-013

14. A safe cast contract

What and why
Spend Sap on an accepted cast; telegraph for 900 ms, then apply at a legal turn boundary.
Resolve own chain, apply due effects, offset attacks, shield, land up to 30, then spawn.
Freeze target choices at commit; fizzle illegal targets safely. Never overwrite a falling pair.

Where on screen Power button, target rail warning and a dedicated pending cast queue.

Effort L | Priority 2 | PNL-014

15. Sap that cannot farm itself

What and why
At natural chain completion grant 0.7 per coloured pod plus 2 per chain link, once per chain.
Add 0.5 per ordinary Chaff landed, capped at 6 Sap per turn; no Sap from power effects.
Clamp to 100. A three chain of twelve pods grants 14.4 Sap; see power_resolution_contract.md.

Where on screen Sap rail with a numeric value, cost notch and an accessible Ready label.

Effort M | Priority 2 | PNL-015

16. Deep Roots

What and why
Keeper | 40 Sap | 12 s cooldown. Prototype balance, not final.
Remove up to 12 own Chaff, highest first, then left to right. Resolve without new rewards.
Root claws lift grey husks; three dry plucks. Replaces unbounded Solvent.

Where on screen Powers loadout, keeper portrait and target warning tray.

Effort M | Priority 2 | PNL-016

17. Bloom Surge

What and why
Keeper | 45 Sap | 12 s cooldown. Prototype balance, not final.
Next natural clear adds 50 percent to forwarded attack, capped at 12 extra Chaff.
A flower blooms around the attack icon; rising synth triad. Expires after three placements.

Where on screen Powers loadout, keeper portrait and target warning tray.

Effort M | Priority 2 | PNL-017

18. Greenhouse Glass

What and why
Keeper | 35 Sap | 10 s cooldown. Prototype balance, not final.
Block at most 12 Chaff in the next landing batch; retain surplus in normal order.
A glass dome icon loses one segment; soft bell. Expires after three placements.

Where on screen Powers loadout, keeper portrait and target warning tray.

Effort M | Priority 2 | PNL-018

19. Aphid Rain

What and why
Aphid Swarm | 25 Sap | 9 s cooldown. Prototype balance, not final.
Queue 6 ordinary Chaff. It can be countered, shielded and capped like any attack.
Six tiny husk arrows fall into the warning tray; dry shaker. No forced immediate drop.

Where on screen Powers loadout, aphid swarm portrait and target warning tray.

Effort M | Priority 2 | PNL-019

20. Scratch Back

What and why
Gnat King Cole | 35 Sap | 11 s cooldown. Prototype balance, not final.
Cancel up to 8 own pending Chaff. Only if 8 were cancelled, queue 2 at the rival.
A record reverses beside two chevrons; record scratch. Replaces Dizzy Buzz.

Where on screen Powers loadout, gnat king cole portrait and target warning tray.

Effort M | Priority 2 | PNL-020

21. Wing Sweep

What and why
Mabel Cabbagewing | 35 Sap | 11 s cooldown. Prototype balance, not final.
Clear up to 6 own Chaff from the two tallest columns, highest first; resolve without new rewards.
Two wing arcs sweep out rubble; soft flutter. Replaces the sudden fall speed increase.

Where on screen Powers loadout, mabel cabbagewing portrait and target warning tray.

Effort M | Priority 2 | PNL-021

22. Slime Coat

What and why
Sir Reginald Slugmore | 40 Sap | 12 s cooldown. Prototype balance, not final.
For three placements reduce each ordinary landing batch by 2, with 6 total prevention.
Three shield dots melt one at a time; low rubber squeak. Never recolour the foundation.

Where on screen Powers loadout, sir reginald slugmore portrait and target warning tray.

Effort M | Priority 2 | PNL-022

23. Sever

What and why
Chompers Cutworm | 35 Sap | 11 s cooldown. Prototype balance, not final.
Remove up to 6 own Chaff in the column with the most Chaff; ties choose leftmost.
Scissor shaped jaws cut a grey stack icon; crisp snip. Replaces blocked rotation.

Where on screen Powers loadout, chompers cutworm portrait and target warning tray.

Effort M | Priority 2 | PNL-023

24. Carapace

What and why
Baron Von Beetle | 50 Sap | 14 s cooldown. Prototype balance, not final.
Remove up to 6 own Chaff and block 6 from the next landing batch within three placements.
A shell closes around six grey beads; muted metal clack. Both benefits are capped.

Where on screen Powers loadout, baron von beetle portrait and target warning tray.

Effort M | Priority 2 | PNL-024

25. Shell Shuffle

What and why
Escargeddon | 40 Sap | 12 s cooldown. Prototype balance, not final.
Swap the order of the next two unspawned pairs on your own board once.
Two visible pair cards exchange around a shell; tape rewind. Replaces Slime Spiral.

Where on screen Powers loadout, escargeddon portrait and target warning tray.

Effort M | Priority 2 | PNL-025

26. Crow Cache

What and why
Cawlin The Crow | 40 Sap | 12 s cooldown. Prototype balance, not final.
Store up to 6 own pending Chaff; release it as a cancelable attack after two own locks.
A feathered pouch fills, with two count dots; two caws. Colours and previews stay visible.

Where on screen Powers loadout, cawlin the crow portrait and target warning tray.

Effort M | Priority 2 | PNL-026

27. Undermine

What and why
Gustavo The Gopher | 45 Sap | 12 s cooldown. Prototype balance, not final.
Remove up to 8 own Chaff, lowest first; settle and scan at the safe boundary.
A tunnel scoops grey rubble from below; short rumble. No opponent column lifting.

Where on screen Powers loadout, gustavo the gopher portrait and target warning tray.

Effort M | Priority 2 | PNL-027

28. Hooffall

What and why
Duchess Dapple | 45 Sap | 12 s cooldown. Prototype balance, not final.
Queue two cancelable packets of 4 Chaff, one now and one after the next own lock.
Two hoof stamps show two packet counts; paired kicks. Never hard drop the rival pair.

Where on screen Powers loadout, duchess dapple portrait and target warning tray.

Effort M | Priority 2 | PNL-028

29. Vine Buffer

What and why
Lady Bindweed | 50 Sap | 14 s cooldown. Prototype balance, not final.
Defer up to 12 own pending Chaff for two own locks; it remains offsettable while deferred.
A vine wraps the queue with two count dots; elastic pluck. It delays rather than deletes.

Where on screen Powers loadout, lady bindweed portrait and target warning tray.

Effort M | Priority 2 | PNL-029

30. Spore Bloom

What and why
Miss Mildew | 50 Sap | 14 s cooldown. Prototype balance, not final.
Queue 4 Chaff, plus 1 per 3 settled rival Chaff at commit, capped at 8 total.
Four to eight spore dots enter the warning tray; dusty hiss. No writes into empty cells.

Where on screen Powers loadout, miss mildew portrait and target warning tray.

Effort M | Priority 2 | PNL-030

31. Grey Tide

What and why
Baron Greymould | 70 Sap | 18 s cooldown. Prototype balance, not final.
Queue 12 cancelable Chaff after a 1200 ms tell. No haste or additional handicap.
A crown above two row icons; two low toms. The threat is large, visible and answerable.

Where on screen Powers loadout, baron greymould portrait and target warning tray.

Effort M | Priority 2 | PNL-031

32. Iai Strike

What and why
Ronin Hare | 65 Sap | 16 s cooldown. Prototype balance, not final.
For two own locks parry the next landing batch up to 10; queue exactly that amount back.
A sheath glints with two count dots; one clean swish. Reflected packets cannot reflect again.

Where on screen Powers loadout, ronin hare portrait and target warning tray.

Effort M | Priority 2 | PNL-032

33. Ship powers in a tested order

What and why
First Deep Roots and Aphid Rain; next the defensive set; finally delayed and reflected packets.
Add per packet origin, reflect count and sequence IDs. Give AI the same costs and boundaries.
Keep Powers out of ranked matchmaking until every effect passes replay and overflow tests.

Where on screen Powers Campaign button replaces the hidden URL switch after the complete slice is tested.

Effort L | Priority 2 | PNL-033

Modes and progression

34. A daily worth opening

What and why
Daily Mix gives everyone one seeded three minute score challenge with fixed Standard rules.
Allow unlimited practice; rank the first completed attempt separately from personal best.
Use a server UTC day for online ranking, show local reset time, and offer a seven day archive.

Where on screen Title > Daily Mix card; result shows score, best chain and tomorrow time.

Effort L | Priority 2 | PNL-034

35. Authored puzzles with real solutions

What and why
Launch 36 puzzles in six lessons: four, stairs, sandwich, split, Chaff and counter planning.
Each stores a grid, fixed pair queue, move budget and verified solution; allow undo and reset.
Solve every puzzle in the real engine. Offer a first step hint before showing the whole answer.

Where on screen Title > Puzzle Garden; tutorial opens the first six lessons.

Effort L | Priority 2 | PNL-035

36. Reward mastery without chores

What and why
Each first campaign win recruits that pest and unlocks its dance on a collection floor.
Optional medals reward a three chain, an offset and an All Clear; no streak loss or energy.
Keeper outfits and tags use in game achievements. Keep the existing Sunbeam cap separate.

Where on screen Campaign cards, results and new Crew collection screen.

Effort M | Priority 2 | PNL-036

37. A neighbourhood with a reason to dance

What and why
The Keeper restores a silent garden block by inviting rival crews back onto its dance floor.
Pests 1 to 3 teach expression; 4 to 6 test craft; 7 to 11 reclaim neighbourhood locations.
Mildew reveals the silence spreading; Greymould wants one grey rhythm; Hare tests mastery.

Where on screen One skippable illustrated intro card before each fight; dialogue in runtime text.

Effort M | Priority 2 | PNL-037

38. Short story beats and a proper ending

What and why
Use two lines per card and a clear Play button. First loss offers one practical hint.
After Greymould, the whole crew restores colour; the Hare arrives with an optional invitation.
Beating the Hare opens practice against any rival and a replay of the restored dance floor.

Where on screen Campaign intro, first defeat card, keeper_ending and secret invitation.

Effort M | Priority 2 | PNL-038

39. Training that teaches the satisfying part

What and why
Add freeze time, one step advance, fixed seed restart and a visible chain explanation.
After a loss, offer Replay last placement with undo in practice, without altering the record.
Preserve Solo Endless; add a relaxed untimed practice option instead of more grind.

Where on screen Title > Practice and results > Try that placement.

Effort L | Priority 2 | PNL-039

40. Price around a complete experience

What and why
My design judgment: 99 cents is reasonable after input, save, readability and crash gates pass.
A 2.99 to 3.99 edition needs polished campaign, 36 puzzles, daily archive and dance collection.
These are pricing hypotheses, not sales forecasts. Do not raise price for unfinished Powers.

Where on screen Store promise and launch scope; prioritize offline value before ranked online scope.

Effort S | Priority 2 | PNL-040

Versus and social

41. Rematch without rebuilding the room

What and why
Add Rematch, Leave and a visible opponent ready state; retain the room and create a new match ID.
Use a fresh shared seed after both agree. Reset event counters and reject stale match packets.
Guest room play should not require a profile; offer sign in only for persistent online records.

Where on screen Versus result screen, where the live code currently hides Again.

Effort L | Priority 2 | PNL-041

42. Quick match that tells the truth

What and why
Use one Standard casual queue first, with region and coarse skill bands that widen gradually.
After 30 seconds offer a clearly labelled bot or room code; never disguise a bot as a person.
Resolve simultaneous top outs as draws and use server validated results for ranked rewards.

Where on screen Versus lobby > Quick match; queue status and result record.

Effort L | Priority 3 | PNL-042

43. Two people on one phone, comfortably

What and why
Portrait defaults to pass and play: same seed and time budget, alternate three rounds.
Real time shared phone play is an optional landscape table mode with separate pointer ownership.
Two full boards plus controls do not fit comfortably in portrait; do not shrink targets to force it.

Where on screen Versus > On this phone; turn handoff screen and optional table layout.

Effort L | Priority 3 | PNL-043

44. Make online attacks durable

What and why
The source publishes only the latest eight attack events with client authoritative counts.
Add ordered acknowledgements, replayable event history and match scoped validation.
Test duplicate, missing and reordered packets, reconnects and stale room snapshots.

Where on screen Online transport, pending Chaff strip and connection status beside rival board.

Effort L | Priority 1 | PNL-044

45. Share a result rather than an account

What and why
Generate a local result card with ruleset, best chain and a challenge seed, without personal data.
Keep mute, leave and a simple report control for online play; avoid open chat at launch.
One tap copies a room code; display the code as real selectable text over the cassette art.

Where on screen Results Share and the existing cassette room card in Versus lobby.

Effort M | Priority 3 | PNL-045

The dance battle

46. Let the chain conduct the dancers

What and why
At chain 2 play the six frame signature loop once; at chain 3 to 4 add a beat accent.
At chain 5 or more hold the signature freeze for 240 ms and let the crowd answer.
Never cover the player grid. Queue reactions by resolve event, not by redraw frame.

Where on screen Rival portrait rail and a small Keeper stage below the rival board.

Effort M | Priority 2 | PNL-046

47. A clear reaction priority

What and why
Priority: match result, hit, danger, chain dance, idle. Return to the current state after hit.
Enter danger at spawn column height 10; exit at 8 to avoid flicker. React to landed Chaff only.
Use a 180 ms hit and a slow danger pose; freeze motion instead of flashing in reduced motion.

Where on screen Character rail; hit and danger sprites use the same anchor and scale as dance frames.

Effort M | Priority 2 | PNL-047

48. Use each signature, not the same bounce

What and why
Aphids wave, Cole scratches, Mabel backspins, Slugmore robots, Chompers worms, Beetle flares.
Escargeddon shell spins; Crow freezes; Gopher pops up; Dapple uprocks; Bindweed pretzels.
Mildew moonwalks; Greymould freezes; Hare headspins; Keeper top rocks then plants a seed.

Where on screen Crew collection and match portraits; named animation map in extras.

Effort L | Priority 2 | PNL-048

49. Sound with space for thought

What and why
Give movement a soft tick, rotation a short click, landing a woody thump and Chaff a dry rattle.
Raise chain pitches through a short scale; distinguish rival timbre and duck music during cues.
Separate music, effects and haptics toggles. Preview each cue; keep all information visible.

Where on screen Settings audio controls and both boards; optional eighties beat remains optional.

Effort M | Priority 2 | PNL-049

Difficulty

50. The first three rivals are lessons

What and why
Start with the current 0.80, 0.95 and 1.15 cells per second; do not speed them up blindly.
Aphids teach four and landing; Cole teaches a two chain; Mabel teaches a six Chaff offset.
Use guided restarts and optional prompts. Target first three win rates of 80, 70 and 60 percent.

Where on screen Campaign first three stages; onboarding prompts stop after the player demonstrates the skill.

Effort M | Priority 1 | PNL-050

51. Make the middle ladder readable

What and why
Teach the fifth colour before stage 9; preview clutter before stages 7, 9 and 11.
Use identical seeded starting patterns for both boards, not merely the same number of pods.
Scale AI planning and cadence gradually. Keep assisted and standard completion badges separate.

Where on screen Campaign cards, stage preview and seeded board setup.

Effort M | Priority 2 | PNL-051

52. The secret should punish decisions, not thumbs

What and why
The brief reports 8 percent wins; this review did not reproduce that measurement.
Trial Hare at 8.5 cells per second, 300 ms reaction and 1.0 attack multiplier, then measure.
Target 15 percent expert match wins in best of three; do not confuse round and match win rates.

Where on screen Ronin Hare challenge; preserve access and instant retry after every defeat.

Effort M | Priority 2 | PNL-052

53. A practical balance experiment

What and why
Use fixed seed suites for regression and at least 200 expert rounds for a first difficulty estimate.
Report player cohort, rules, controls, round count, match count and confidence intervals.
Change one knob at a time. Bot sweeps check consistency; human testing decides comfort.

Where on screen Internal balance report, never shown as a claim on the store page.

Effort M | Priority 1 | PNL-053

Art and look

54. The well owns the contrast

What and why
Keep the painted alley but place an opaque near black well behind the new speaker frame.
Keep particles outside the active piece and preview; the character accent belongs in the rail.
The first page is a visual proposal, not a screenshot or proof of implementation.

Where on screen Match background, player frame and right rail.

Effort M | Priority 1 | PNL-054

55. Pods that read without colour

What and why
Use crimson diamond, sprout triangle, dewdrop circle, sunbean square and nightshade star.
Keep the code drawn faces; place each marking low enough that eyes never cover it.
Check at 38, 40 and 44 px in greyscale. Retain the existing tag shapes option as a stronger overlay.

Where on screen Player cells, next queue, ghost guide and rival mini board.

Effort M | Priority 1 | PNL-055

56. Connect groups without muddying them

What and why
Place horizontal and vertical joins behind equal coloured orthogonal neighbours only.
Clip joins to cell bounds and never connect through Chaff, diagonals or empty cells.
Use the same shape cues on all pod bodies; connected groups must retain individual cell edges.

Where on screen Both wells; ten colour specific link files under pods by colour subfolder.

Effort M | Priority 2 | PNL-056

57. Asset integration is explicit

What and why
This art pass includes 111 of 216 requested files. The manifest lists all 105 missing files.
Map the new underscore filenames to the existing hyphenated runtime IDs before loading.
Preload only current opponents; retain the procedural fallback and unload unused animation sets.

Where on screen Asset loader, portrait state machine and mobile memory budget.

Effort M | Priority 1 | PNL-057

58. Keep the visual language accessible

What and why
Use text plus shape for tier, incoming attacks and power readiness; never colour alone.
All player copy avoids dashes and exclamation marks. Studio credit is Sky Wolf Studio.
Offer reduced motion, no camera shake, larger text and separate haptics; preserve sound cues visually.

Where on screen Title, settings, HYPE tier plate, power warnings, results and story cards.

Effort M | Priority 1 | PNL-058

59. A launch gate that matches the promise

What and why
Check 412 x 915, 375 wide and 360 wide, expanded browser bars, safe areas and 200 percent text.
On target Android hardware measure 60 fps, input to display under 50 ms at the 95th percentile.
Play offline after install, interrupt and restore a run, and finish a full ladder before pricing up.

Where on screen Physical device test plan; the review only verifies source and analytical viewport geometry.

Effort M | Priority 1 | PNL-059

Bugs you saw

60. HYPE drains in milliseconds

What and why
Isolated live source test: HYPE 80 reaches 0 in four 16 ms ticks, only 64 ms.
Change dt * 1.3 to dt / 1000 * 1.3 for the intended 1.3 units per second.
Proposed fix passed the isolated test at 79.9168. Full gameplay integration remains untested.

Where on screen Match right edge HYPE meter; tickField.

Effort S | Priority 1 | PNL-060

61. The rival chain glow vanishes immediately

What and why
Isolated source test: chainGlow 1 becomes 0 after one 16 ms tick.
Change dt * 1.1 to dt / 1000 * 1.1; the corrected first value is 0.9824.
Keep this timer independent from the pop phase so the opponent attack remains readable.

Where on screen Rival frame flare; tickField and render.

Effort S | Priority 1 | PNL-061

62. All Clear is spent without a clear

What and why
Isolated source test: stored bonus plus a zero clear placement sends 30 Chaff and clears the flag.
Consume allBonus only when R.chain > 0, then detect whether this turn earns a new bonus.
The two turn proposed fix passes: retain on no clear, spend once on the next natural clear.

Where on screen How to play promise, finishResolve and All Clear indicator.

Effort S | Priority 1 | PNL-062

63. Touch cancellation can become a move

What and why
The same dragEnd handles pointerup and pointercancel; a cancelled short touch rotates.
The isolated test reproduced one unwanted rotation. Cancel must clear state and softHold only.
Also clear holds if a piece locks mid drag; a gesture must never leak into the next piece.

Where on screen Board gesture handling, OS gestures, notifications and interrupted touches.

Effort S | Priority 1 | PNL-063

64. Solvent can mutate an unsafe board

What and why
firePower has no playable or phase guard and its apply function immediately changes the grid.
Queuing a cleanse at the legal boundary avoids collision and stale resolve masks.
This is a source identified risk, not a reproduced crash. Replace silent catch with a logged failure.

Where on screen Hidden Powers mode, b-power and firePower.

Effort M | Priority 1 | PNL-064

65. Small text and switches shrink below the brief

What and why
The fixed 540 px stage scales to 0.667 at 360 wide. An 11 px label becomes 7.33 px.
The settings switch becomes 42.67 x 24 px; the title back button is 32 px tall. Enlarge both.
The measured play tray passes the 48 px width rule; do not shrink those controls to add more.

Where on screen Settings toggle rows, title footer, NEXT label and rival labels.

Effort M | Priority 1 | PNL-065

66. The next window art and pods use different boxes

What and why
In render, next-window is fit to a box centred at nx + 26, ny + 43 with size 62 x 98.
The pair is drawn separately at nx + 11 and can exceed the narrow contained art aperture.
Derive both from one preview rectangle. This is a source geometry finding requiring device QA.

Where on screen Match right rail > next pair window.

Effort S | Priority 1 | PNL-066

67. The fixed input delay is longer than its label

What and why
hold waits 105 ms, starts an interval, then waits another 33 ms before the first repeat.
Execute the first repeat at the end of the initial delay, then schedule the repeat interval.
Use the shared simulation input scheduler so browser timer throttling cannot change cadence.

Where on screen Left and right hold handlers in the play tray.

Effort S | Priority 1 | PNL-067

68. Existing polish should be retained

What and why
The celebration already supports skip, the top buttons extend their hit areas, and stage fit uses visualViewport.
The live source has a procedural art fallback and room seed support. Preserve these behaviours.
No authenticated online match, real phone session or complete campaign playthrough was claimed.

Where on screen Outro tap handler, fit, asset loader and Versus seed setup.

Effort S | Priority 1 | PNL-068

Implementation JSON

The array below is also included as extras/directions.json in the asset pack.


```
[
  {"id": "PNL-001", "title": "Fix the feel before expanding the game", "group": "Core gameplay",
   "effort": "M", "priority": 1,
   "summary": "First ship the verified fixes, dependable input and readable boards. Then add progression."},
  {"id": "PNL-002", "title": "Separate rules from controls", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Keep the current countering rules as Standard; offer Retro without countering or All Clear."},
  {"id": "PNL-003", "title": "A small landing window", "group": "Core gameplay", "effort": "M", "priority": 1,
   "summary": "Start a 300 ms grounded timer on first contact, with at most two successful move resets."},
  {"id": "PNL-004", "title": "Deliberate rotation and quick turn", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Test rotation at both walls, on ledges, beneath roofs and inside a one column shaft."},
  {"id": "PNL-005", "title": "One predictable repeat scheduler", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Keep 105 ms delay and 33 ms repeat as the Fast preset; offer 160 ms and 50 ms for Relaxed."},
  {"id": "PNL-006", "title": "Separate movement from drop intent", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Default to buttons and soft drop. Make the upward hard drop flick an explicit optional setting."},
  {"id": "PNL-007", "title": "Bring back a visible soft drop", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "The live tray has five buttons and no c-down element, although a soft hold handler exists."},
  {"id": "PNL-008", "title": "Use the phone height without shrinking text", "group": "Core gameplay",
   "effort": "M", "priority": 1,
   "summary": "At 412 wide, propose a 264 x 528 player grid and a 96 x 192 rival grid in a separate rail."},
  {"id": "PNL-009", "title": "Keep the landing guide accurate", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Show both destination cells including split landings; a faint connector marks the active pivot."},
  {"id": "PNL-010", "title": "A readable chain rhythm", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Tuning start: 80 ms landing squash, 230 ms pop, 110 to 180 ms settle, then the next scan."},
  {"id": "PNL-011", "title": "Make every attack explainable", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Display queued Chaff as a number plus row icons, with a separate cancelled amount."},
  {"id": "PNL-012", "title": "Offline comfort and fair interruption", "group": "Core gameplay", "effort": "M",
   "priority": 1,
   "summary": "Pause Solo and Campaign when the app hides; save board, queue, seed and clocks atomically."},
  {"id": "PNL-013", "title": "Replace control sabotage with board tactics", "group": "Powers", "effort": "L",
   "priority": 2,
   "summary": "Cut input reversal, rotation lock, forced drop, hidden colours and full control freeze."},
  {"id": "PNL-014", "title": "A safe cast contract", "group": "Powers", "effort": "L", "priority": 2,
   "summary": "Spend Sap on an accepted cast; telegraph for 900 ms, then apply at a legal turn boundary."},
  {"id": "PNL-015", "title": "Sap that cannot farm itself", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "At natural chain completion grant 0.7 per coloured pod plus 2 per chain link, once per chain."},
  {"id": "PNL-016", "title": "Deep Roots", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Remove up to 12 own Chaff, highest first, then left to right. Resolve without new rewards."},
  {"id": "PNL-017", "title": "Bloom Surge", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Next natural clear adds 50 percent to forwarded attack, capped at 12 extra Chaff."},
  {"id": "PNL-018", "title": "Greenhouse Glass", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Block at most 12 Chaff in the next landing batch; retain surplus in normal order."},
  {"id": "PNL-019", "title": "Aphid Rain", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Queue 6 ordinary Chaff. It can be countered, shielded and capped like any attack."},
  {"id": "PNL-020", "title": "Scratch Back", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Cancel up to 8 own pending Chaff. Only if 8 were cancelled, queue 2 at the rival."},
  {"id": "PNL-021", "title": "Wing Sweep", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Clear up to 6 own Chaff from the two tallest columns, highest first; resolve without new rewards."},
  {"id": "PNL-022", "title": "Slime Coat", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "For three placements reduce each ordinary landing batch by 2, with 6 total prevention."},
  {"id": "PNL-023", "title": "Sever", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Remove up to 6 own Chaff in the column with the most Chaff; ties choose leftmost."},
  {"id": "PNL-024", "title": "Carapace", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Remove up to 6 own Chaff and block 6 from the next landing batch within three placements."},
  {"id": "PNL-025", "title": "Shell Shuffle", "group": "Powers", "effort": "M", "priority": 2,
   "summary": "Swap the order of the next two unspawned pairs on your own board once."},
  {"id": "PNL-026", "title": "Crow Cache", "group": "Powers", "effort": "M", "priority": 2,
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 "summary": "Store up to 6 own pending Chaff; release it as a cancelable attack after two own locks."},
{"id": "PNL-027", "title": "Undermine", "group": "Powers", "effort": "M", "priority": 2,
 "summary": "Remove up to 8 own Chaff, lowest first; settle and scan at the safe boundary."},
{"id": "PNL-028", "title": "Hooffall", "group": "Powers", "effort": "M", "priority": 2,
 "summary": "Queue two cancelable packets of 4 Chaff, one now and one after the next own lock."},
{"id": "PNL-029", "title": "Vine Buffer", "group": "Powers", "effort": "M", "priority": 2,
 "summary": "Defer up to 12 own pending Chaff for two own locks; it remains offsettable while deferred."},
{"id": "PNL-030", "title": "Spore Bloom", "group": "Powers", "effort": "M", "priority": 2,
 "summary": "Queue 4 Chaff, plus 1 per 3 settled rival Chaff at commit, capped at 8 total."},
{"id": "PNL-031", "title": "Grey Tide", "group": "Powers", "effort": "M", "priority": 2,
 "summary": "Queue 12 cancelable Chaff after a 1200 ms tell. No haste or additional handicap."},
{"id": "PNL-032", "title": "Iai Strike", "group": "Powers", "effort": "M", "priority": 2,
 "summary": "For two own locks parry the next landing batch up to 10; queue exactly that amount back."},
{"id": "PNL-033", "title": "Ship powers in a tested order", "group": "Powers", "effort": "L", "priority": 2,
 "summary": "First Deep Roots and Aphid Rain; next the defensive set; finally delayed and reflected packets."},
{"id": "PNL-034", "title": "A daily worth opening", "group": "Modes and progression", "effort": "L",
 "priority": 2,
 "summary": "Daily Mix gives everyone one seeded three minute score challenge with fixed Standard rules."},
{"id": "PNL-035", "title": "Authored puzzles with real solutions", "group": "Modes and progression",
 "effort": "L", "priority": 2,
 "summary": "Launch 36 puzzles in six lessons: four, stairs, sandwich, split, Chaff and counter planning."},
{"id": "PNL-036", "title": "Reward mastery without chores", "group": "Modes and progression", "effort": "M",
 "priority": 2,
 "summary": "Each first campaign win recruits that pest and unlocks its dance on a collection floor."},
{"id": "PNL-037", "title": "A neighbourhood with a reason to dance", "group": "Modes and progression",
 "effort": "M", "priority": 2,
 "summary": "The Keeper restores a silent garden block by inviting rival crews back onto its dance floor."},
{"id": "PNL-038", "title": "Short story beats and a proper ending", "group": "Modes and progression",
 "effort": "M", "priority": 2,
 "summary": "Use two lines per card and a clear Play button. First loss offers one practical hint."},
{"id": "PNL-039", "title": "Training that teaches the satisfying part", "group": "Modes and progression",
 "effort": "L", "priority": 2,
 "summary": "Add freeze time, one step advance, fixed seed restart and a visible chain explanation."},
{"id": "PNL-040", "title": "Price around a complete experience", "group": "Modes and progression",
 "effort": "S", "priority": 2,
 "summary": "My design judgment: 99 cents is reasonable after input, save, readability and crash gates pass."},
{"id": "PNL-041", "title": "Rematch without rebuilding the room", "group": "Versus and social",
 "effort": "L", "priority": 2,
 "summary": "Add Rematch, Leave and a visible opponent ready state; retain the room and create a new match ID."},
{"id": "PNL-042", "title": "Quick match that tells the truth", "group": "Versus and social", "effort": "L",
 "priority": 3,
 "summary": "Use one Standard casual queue first, with region and coarse skill bands that widen gradually."},
{"id": "PNL-043", "title": "Two people on one phone, comfortably", "group": "Versus and social",
 "effort": "L", "priority": 3,
 "summary": "Portrait defaults to pass and play: same seed and time budget, alternate three rounds."},
{"id": "PNL-044", "title": "Make online attacks durable", "group": "Versus and social", "effort": "L",
 "priority": 1,
 "summary": "The source publishes only the latest eight attack events with client authoritative counts."},
{"id": "PNL-045", "title": "Share a result rather than an account", "group": "Versus and social",
 "effort": "M", "priority": 3,
 "summary": "Generate a local result card with ruleset, best chain and a challenge seed, without personal data."},
{"id": "PNL-046", "title": "Let the chain conduct the dancers", "group": "The dance battle", "effort": "M",
 "priority": 2,
 "summary": "At chain 2 play the six frame signature loop once; at chain 3 to 4 add a beat accent."},
{"id": "PNL-047", "title": "A clear reaction priority", "group": "The dance battle", "effort": "M",
 "priority": 2,
 "summary": "Priority: match result, hit, danger, chain dance, idle. Return to the current state after hit."},
{"id": "PNL-048", "title": "Use each signature, not the same bounce", "group": "The dance battle",
 "effort": "L", "priority": 2,
 "summary": "Aphids wave, Cole scratches, Mabel backspins, Slugmore robots, Chompers worms, Beetle flares."},
{"id": "PNL-049", "title": "Sound with space for thought", "group": "The dance battle", "effort": "M",
 "priority": 2,
 "summary": "Give movement a soft tick, rotation a short click, landing a woody thump and Chaff a dry rattle."},
{"id": "PNL-050", "title": "The first three rivals are lessons", "group": "Difficulty", "effort": "M",
 "priority": 1,
 "summary": "Start with the current 0.80, 0.95 and 1.15 cells per second; do not speed them up blindly."},
{"id": "PNL-051", "title": "Make the middle ladder readable", "group": "Difficulty", "effort": "M",
 "priority": 2,
 "summary": "Teach the fifth colour before stage 9; preview clutter before stages 7, 9 and 11."},
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  {"id": "PNL-052", "title": "The secret should punish decisions, not thumbs", "group": "Difficulty",
   "effort": "M", "priority": 2,
   "summary": "The brief reports 8 percent wins; this review did not reproduce that measurement."},
  {"id": "PNL-053", "title": "A practical balance experiment", "group": "Difficulty", "effort": "M",
   "priority": 1,
   "summary": "Use fixed seed suites for regression and at least 200 expert rounds for a first difficulty estimate."},
  {"id": "PNL-054", "title": "The well owns the contrast", "group": "Art and look", "effort": "M",
   "priority": 1,
   "summary": "Keep the painted alley but place an opaque near black well behind the new speaker frame."},
  {"id": "PNL-055", "title": "Pods that read without colour", "group": "Art and look", "effort": "M",
   "priority": 1,
   "summary": "Use crimson diamond, sprout triangle, dewdrop circle, sunbean square and nightshade star."},
  {"id": "PNL-056", "title": "Connect groups without muddying them", "group": "Art and look", "effort": "M",
   "priority": 2,
   "summary": "Place horizontal and vertical joins behind equal coloured orthogonal neighbours only."},
  {"id": "PNL-057", "title": "Asset integration is explicit", "group": "Art and look", "effort": "M",
   "priority": 1,
   "summary": "This art pass includes 111 of 216 requested files. The manifest lists all 105 missing files."},
  {"id": "PNL-058", "title": "Keep the visual language accessible", "group": "Art and look", "effort": "M",
   "priority": 1,
   "summary": "Use text plus shape for tier, incoming attacks and power readiness; never colour alone."},
  {"id": "PNL-059", "title": "A launch gate that matches the promise", "group": "Art and look", "effort": "M",
   "priority": 1,
   "summary": "Check 412 x 915, 375 wide and 360 wide, expanded browser bars, safe areas and 200 percent text."},
  {"id": "PNL-060", "title": "HYPE drains in milliseconds", "group": "Bugs you saw", "effort": "S",
   "priority": 1, "summary": "Isolated live source test: HYPE 80 reaches 0 in four 16 ms ticks, only 64 ms."},
  {"id": "PNL-061", "title": "The rival chain glow vanishes immediately", "group": "Bugs you saw",
   "effort": "S", "priority": 1,
   "summary": "Isolated source test: chainGlow 1 becomes 0 after one 16 ms tick."},
  {"id": "PNL-062", "title": "All Clear is spent without a clear", "group": "Bugs you saw", "effort": "S",
   "priority": 1,
   "summary": "Isolated source test: stored bonus plus a zero clear placement sends 30 Chaff and clears the flag."},
  {"id": "PNL-063", "title": "Touch cancellation can become a move", "group": "Bugs you saw", "effort": "S",
   "priority": 1,
   "summary": "The same dragEnd handles pointerup and pointercancel; a cancelled short touch rotates."},
  {"id": "PNL-064", "title": "Solvent can mutate an unsafe board", "group": "Bugs you saw", "effort": "M",
   "priority": 1,
   "summary": "firePower has no playable or phase guard and its apply function immediately changes the grid."},
  {"id": "PNL-065", "title": "Small text and switches shrink below the brief", "group": "Bugs you saw",
   "effort": "M", "priority": 1,
   "summary": "The fixed 540 px stage scales to 0.667 at 360 wide. An 11 px label becomes 7.33 px."},
  {"id": "PNL-066", "title": "The next window art and pods use different boxes", "group": "Bugs you saw",
   "effort": "S", "priority": 1,
   "summary": "In render, next-window is fit to a box centred at nx + 26, ny + 43 with size 62 x 98."},
  {"id": "PNL-067", "title": "The fixed input delay is longer than its label", "group": "Bugs you saw",
   "effort": "S", "priority": 1,
   "summary": "hold waits 105 ms, starts an interval, then waits another 33 ms before the first repeat."},
  {"id": "PNL-068", "title": "Existing polish should be retained", "group": "Bugs you saw", "effort": "S",
   "priority": 1,
   "summary": "The celebration already supports skip, the top buttons extend their hit areas, and stage fit uses visualViewport."}
]
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